Formative Assignment 2: Task Division

Sentiment Analysis: To Vaccinate or Not to Vaccinate (4-person group)

1. Individual responsibilities

Member	Model to train	Report sections	Other responsibilities	Deliverables
Hamse **Data & Baselines Lead**	Model 1: TF-IDF + Logistic Regression Model 2: TF-IDF + Linear SVM (or Naive Bayes)	Introduction Dataset and Exploratory Analysis References (compile at the end)	Build shared preprocessing pipeline Fix train/val/test split and random seed Set up GitHub repo structure	Cleaned data in /data EDA plots (class balance, length, vocabulary, hashtags, noise) Baseline metrics
Nehemie **Recurrent Model Lead**	Model 3: BiLSTM or BiGRU with embeddings	Related Work (RNNs and sequence models for sentiment) Methodology subsection for Model 3	Hyperparameter tuning Send ~10 misclassified tweets to Person 4	Learning curves Confusion matrix Tuning notes
Gabriel **Transformer Lead**	Model 4: Fine-tuned XLM-R, AfroXLMR or BERTweet (Hugging Face)	Related Work (transformers, multilingual and low-resource NLP) Methodology subsection for Model 4	Tuning and validation strategy Send ~10 misclassified tweets to Person 4	Learning curves Confusion matrix and ROC curves Tuning notes
Mahammat **CNN Model & Analysis Lead**	Model 5: TextCNN (or BiLSTM + attention)	Results and Discussion Error Analysis and Limitations Conclusion and Future Work Evaluation metric justification	Build five-model comparison table Select and analyse misclassified examples Send ~10 misclassified tweets from own model too	Comparison table and figures Error analysis Metric justification (macro-F1, accuracy)